

Lecture 8

Bézout coefficients

$$g = sa + tb$$

Last time: Primes, GCD, Euclidean Alg, Bézout's thm.

ex2: $a = 252, b = 198$

$$\gcd(252, 198) = \text{key.}$$
$$\gcd(198, 54) =$$
$$\gcd(54, 36) =$$
$$\gcd(36, 18) =$$
$$18$$

Division

$$\underset{a}{252} = \underset{q}{1} \cdot \underset{b}{198} + \textcircled{54}$$

$$198 = 3 \cdot \textcircled{54} + \textcircled{36}$$

$$\textcircled{54} = 1 \cdot 36 + \textcircled{18}$$

$$36 = 2 \cdot \textcircled{18} + 0$$

$\underset{g}{18}$

Reverse
Substitute
remainders.

$$18 = 54 - 36$$

$$= 54 - (198 - 3 \cdot 54)$$

$$= 4 \cdot 54 - 198$$

$$= 4(252 - 198) - 198$$

$$= \underline{s} \cdot 252 - \underline{t} \cdot 198$$

Remark: It is possible to use the algorithm to prove Bézout's theorem.

(Q) How do you divide modulo m ?

I Inverses Modulo m

II Inverses modulo p , prime

I Defn: Two integers a, b are relatively prime if $\gcd(a, b) = 1$.

Euclid's Lemma: If $\gcd(a, b) = 1$ and $a | bc$ then $a | c$.

Proof: Assume $a, b \in \mathbb{Z}$, $\gcd(a, b) = 1$, $a | bc$.

By Bézout, we can choose $s, t \in \mathbb{Z}$ s.t. $sq + tb = 1$.

Observe that $c = csa + ctb = \underline{csa} + t(\underline{bc})$.

Observe that $a | csa$ and $a | tbc$ (by Lec 6), so
 $a | c$, as desired. $\square$

a, b, c
 s, t
 $a | bc$

Defn: Given $a \in \mathbb{Z}$, $m > 1$ an integer $\bar{a}$ is an inverse
of a modulo m if $a \bar{a} \equiv 1 \pmod{m}$

ex 1: $a = 5$, $m = 7$.

$\bar{a}$ inverse if $5\bar{a} \equiv 1 \pmod{7}$.

Guess: $\bar{a} = 3$: $5 \cdot 3 \equiv 15 \equiv 1 \pmod{7}$.

$$5 \cdot 10 \equiv 5 \cdot 3 \equiv 1 \pmod{7}$$

$\bar{a} = 3, 10, 17, -4, \dots$ are all inverses.

Theorem: If $a \in \mathbb{Z}$, $m > 1$, $\gcd(a, m) = 1$ then a has
a unique inverse $a^{-1} \in \{1, 2, \dots, m-1\}$.

"division"

$$\frac{a}{x} = a \cdot \left(\frac{1}{x}\right)$$
$$x \cdot \frac{1}{x} = 1$$

Proof: Existence: Assume $a \in \mathbb{Z}$, $m > 1$, $\gcd(a, m) = 1$.

By Bézout, we can choose $s, t \in \mathbb{Z}$ s.t.
 $sa + tm = 1$.

$$\begin{aligned}\text{Thus, } 1 &\equiv sa + tm \pmod{m} \\ &\equiv (s \bmod m) a + \underbrace{t \cdot 0}_{\text{remainder}} \pmod{m} \\ &\equiv (s \bmod m) a \pmod{m}.\end{aligned}$$

Thus, $\bar{a} = s \bmod m$ is an inverse of $a \pmod{m}$.

Uniqueness: <HW>

different.

$s \bmod m = \frac{\text{remainder of } s}{\text{divided by } m}$

$a \equiv b \pmod{m}$
 $\updownarrow$
 $a \bmod m = b \bmod m$

□

ex2: $a=5, m=7$.

Step 1: $7 = 1 \cdot 5 + \textcircled{2}$
 $5 = 2 \cdot 2 + \textcircled{1}$
 $2 = 2 \cdot 1 + 0$

EA

Step 2:

Bezout
 $1 = 5 - 2 \cdot 2$
 $= 5 - 2(7 - 5)$
 $= \underset{s}{3} \cdot 5 - \underset{t}{2} \cdot 7$

Step 3
 $a^{-1} = 3 \pmod{7}$
 $= 3.$

ex3: Find x such that $5x \equiv 3 \pmod{7}$.

Multiply both sides by $5^{-1} \equiv 3 \pmod{7}$:

$$x \equiv \underset{\text{defn}}{5^{-1} \cdot 3} \cdot x \equiv 5^{-1} \cdot 3 \equiv 3 \cdot 3 \equiv 2 \pmod{7}.$$

$\therefore 2$ is the unique solution in $\{0, \dots, 6\}$.

Remark: In general, if $\gcd(a, m) = 1$, you can solve $ax \equiv b$ by multiplying both sides by a^{-1} .

Remark: $\gcd(a, m) = 1$ is essential for an inverse to exist.

ex4: $a = 6, m = 4. \quad \gcd(a, m) = 2.$

Suppose $\bar{a}$ inverse exists.

Observe $a\bar{a}$ is even.

Thus $a\bar{a} \pmod{4}$ is even, so $a\bar{a} \not\equiv 1 \pmod{4}$.

Poll: $a, m > 1, \gcd(a, m) = 1, \quad f(x) = (a \cdot x) \pmod{m}.$
 $f: \underbrace{\{0, \dots, m-1\}}_A \rightarrow \underbrace{\{0, \dots, m-1\}}_B$

Proof: (1-1) Assume $x_1, x_2 \in A$.
Assume $f(x_1) = f(x_2), \text{ i.e. } a \cdot x_1 \pmod{m} = a \cdot x_2 \pmod{m}.$

Since $\gcd(a, m) = 1$, a has an inverse $a^{-1} \pmod{m}$.

Multiplying $a \cdot x_1 \equiv a \cdot x_2 \pmod{m}$ by a^{-1} :

$$x_1 \equiv x_2 \pmod{m}.$$

Since $0 \leq x_1, x_2 < m$, $x_1 = x_2$ as desired.

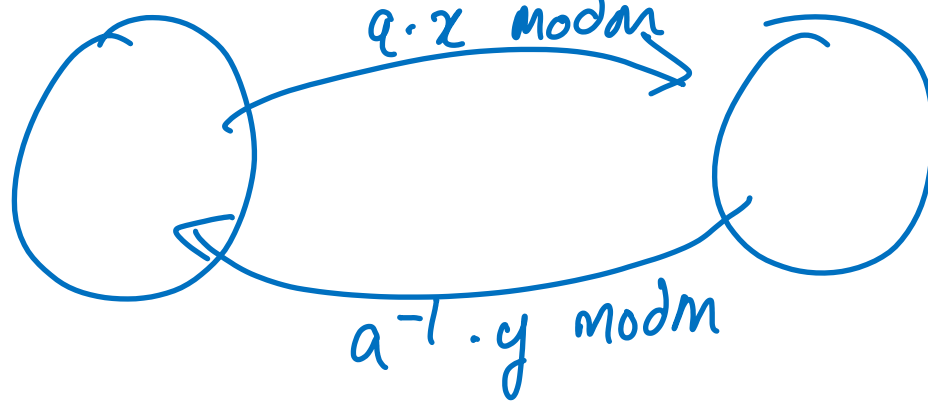
(Onto) Assume $y \in B$. Since $\gcd(a, m) = 1$,
The congruence $ax \equiv y \pmod{m}$ has
a solution, namely $x = a^{-1}y \pmod{m}$.
Thus, $f(x) = a \cdot a^{-1}y \pmod{m}$

$$\begin{aligned} &\equiv aa^{-1}y \pmod{m} \\ &\equiv y \pmod{m}. \end{aligned}$$

Since $f(x), y < m$, we must have
 $f(x) = y$. $\square$

Alternatively.

$g: \{0, \dots, m-1\} \rightarrow \{0, \dots, m-1\}$
by $g(y) = a^{-1}y \pmod{m}$
is an inverse of f .



$$a^{-1} a x = x.$$

II Fermat's Little Theorem (NOT Fermat's Last Thm:
 $\forall a, b, c \in \mathbb{Z}, n > 2 (a^n + b^n \neq c^n)$.
 Ribet '85 + Wiles '94 ~ 300pp)

Theorem: If p is a prime and $a \not\equiv 0 \pmod{p}$ then
 $a^{p-1} \equiv 1 \pmod{p}$.

$\left[\underbrace{a^{p-2}}_{\text{inverse}} \cdot a \equiv 1 \pmod{p} \right]$ "Division reduces to multiplication"

ex 5: $a = 5, p = 7$.

$$\begin{aligned} a^6 &\equiv 5^6 \pmod{7} \\ &\equiv (25)^3 \pmod{7} \\ &\equiv 64 \pmod{7} \\ &\equiv 1 \pmod{7}. \end{aligned}$$

$\therefore a^5 = 5^5$ is an inverse modulo 7.

Remark: This is central to Cryptography (4.7) -

Theorem: If p is a prime and $a \not\equiv 0 \pmod{p}$ then
 $a^{p-1} \equiv 1 \pmod{p}$.

Proof: Assume p is prime and $p \nmid a$.

Consider the numbers

$$(a \cdot 1) (a \cdot 2) \dots (a \cdot (p-1)) \in \mathbb{Z}$$

Consider the remainders

$$r_1 = a \cdot 1 \pmod{p}$$

$$r_2 = a \cdot 2 \pmod{p}$$

$$r_{p-1} = a \cdot (p-1) \pmod{p}$$

Claim 1: $\forall k \in \{1, \dots, p-1\}, r_k \neq 0$.

Claim 2: $\forall k, j \in \{1, \dots, p-1\}, k \neq j \rightarrow r_k \neq r_j$.

Running Example

$$a=5, p=7.$$

$$5 \cdot 1, 5 \cdot 2, 5 \cdot 3, 5 \cdot 4, 5 \cdot 5, 5 \cdot 6$$

$$5, 3, 1, 6, 4, 2$$

No zeros, no repeats.

First:

$$5^6 \cdot 6! \pmod{7}$$

Second:

$$\begin{aligned} 5 \cdot 3 \cdot 1 \cdot 6 \cdot 4 \cdot 2 \\ = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \\ = 6! \end{aligned}$$

We will compute $r_1 r_2 \dots r_{p-1} \bmod p$
in two different ways.

$$5^6 \cdot 6! \equiv 6! \pmod{7}.$$

First: $(r_1 r_2 r_3 \dots r_{p-1}) \bmod p$
 $= (a \cdot 1) (a \cdot 2) \dots (a \cdot (p-1)) \bmod p$
 $\equiv a^{p-1} (p-1)! \pmod{p}.$

(multiply, then
reduce
mod p)

Second: Observe that by Claim 5.1+2, $(r_1 \bmod p)(r_2 \bmod p) \dots (r_{p-1} \bmod p)$
 $r_1 r_2 \dots r_{p-1} \bmod p \equiv (r_1 \bmod p)(r_2 \bmod p) \dots (r_{p-1} \bmod p)$
 $\equiv (p-1)! \pmod{p}$

(reduce, then
multiply)

Thus, $a^{p-1} (p-1)! \equiv (p-1)! \pmod{p}.$

Since $\gcd(p, (p-1)!) = 1$, $(p-1)!$ has an inverse q .

Multiplying both sides by q : $a^{p-1} \equiv 1 \pmod{p}$, as desired. $\square$

Proof of Claim 1: Assume $k \in \{1, \dots, p-1\}$.

Assume for contradiction that $a \cdot k \equiv 0 \pmod{p}$

i.e. $p \mid ka$. By Euclid's lemma, since $\gcd(p, a) = 1$, $p \mid k$ which is absurd since $k < p$. $\square$

p, a
 $\gcd(p, a) = 1$
 k

Proof of Claim 2: Assume $k, j \in \{1, \dots, p-1\}$.

Assume $r_k = r_j$, i.e. $a \cdot k \equiv a \cdot j \pmod{p}$

Since $\gcd(p, a)$, a has an inverse a^{-1} modulo p .

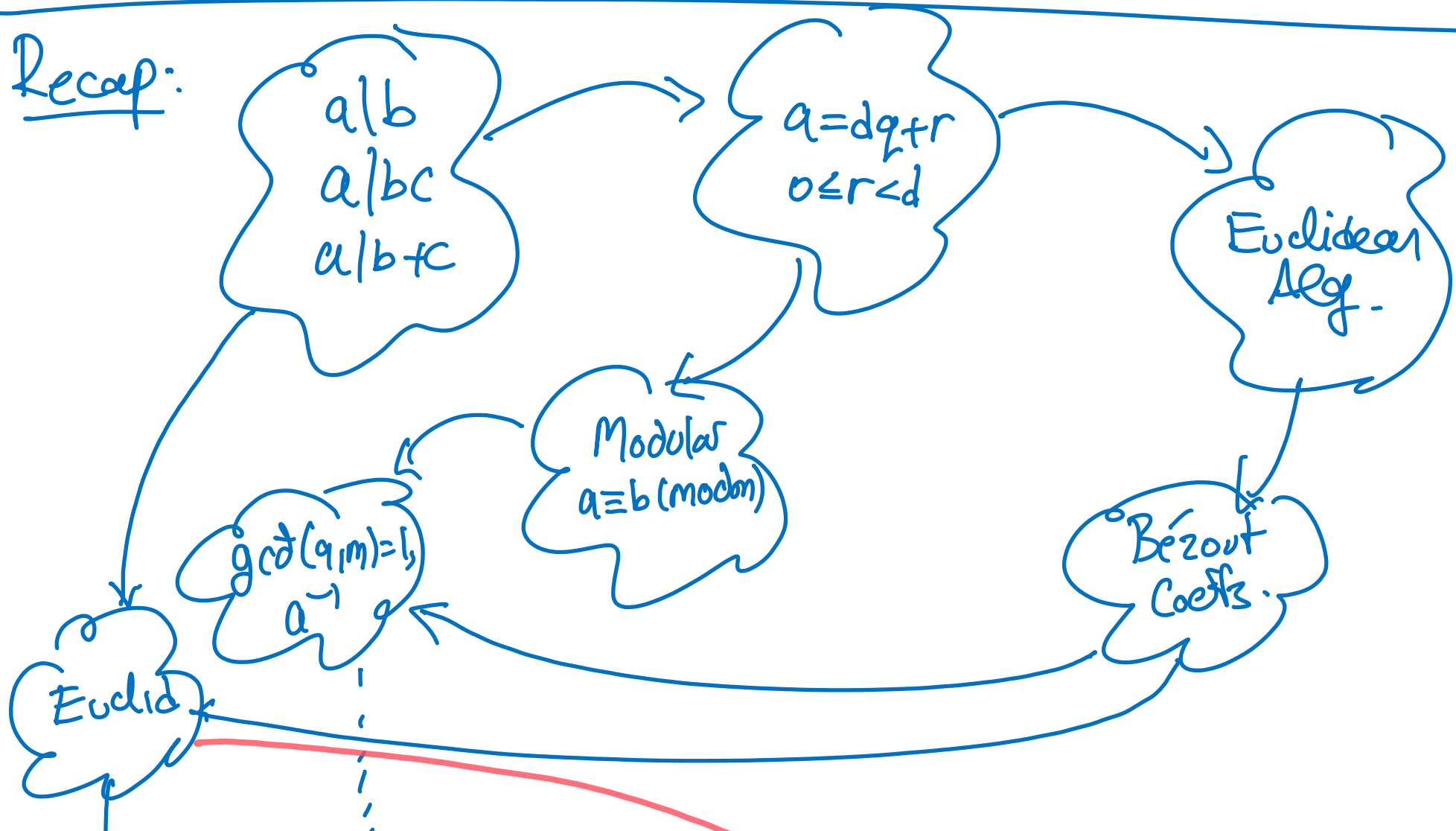
Very similar
to Poll.

$$\therefore a^{-1}ak \equiv a^{-1}aj \pmod{p}$$

i.e. $k \equiv j \pmod{p}$, so

$k=j$ since $0 \leq k, j < p$. $\square$

Recap:



↓
Fermat Little Thm

→
Existence +
Uniqueness of
Prime fact.

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- Midterm next Th. Covers upto Lec 8.
 - Practice Midterms online.
 - No Cheat Sheets.
 - Can cite defs, thms proved in class.